530Q's

Negative Remainder Concept

$$\frac{99}{13}\begin{cases} 8 \text{ (Rem)} \\ -5 \end{cases}$$

D ← 13

-5 or $-5+13=8$

$8 + 5 = 13$

'D'ivisor

13) 99 (7	13) 99 (8
-91	-104
$\boxed{8}$	$\boxed{-5}$
(+ve Rem)	(-ve Rem)

(Rem)

$$\frac{146}{17}\begin{cases} 10 \\ -7 \end{cases}$$

or $-7+17=10$

(Rem)

$$\frac{53}{8}\begin{cases} 5 \\ -3 \end{cases}$$

Rem

$$\frac{85}{11}\begin{cases} 8 \\ -3 \end{cases}$$

$$\frac{-106}{19}\begin{cases} -11 \text{ (Rem)} \\ 8 \end{cases}$$

$$\frac{\overset{2}{73}\times\overset{4}{75}\times\overset{5}{76}\times\overset{8}{79}}{71} = \frac{320}{71} = 36$$

Rem

$$\frac{\overset{-3}{56}\times\overset{-2}{57}\times\overset{-1}{58}\times\overset{5}{64}}{59} = -30 \text{ or } (-30+59) = 29$$

$$\frac{56}{59} \begin{cases} \text{Rem } 56 \\ -3 \end{cases}$$

$$\frac{\underset{-3}{150} \times \underset{-2}{151} \times \underset{5}{158}}{17} =$$

Rem.

$$\frac{\cancel{30}}{17} \to 13 \checkmark$$

$$\frac{150}{17} \to \text{Rem } \cancel{14}, \quad -3$$

Rem

$$\frac{\cancel{95}}{\cancel{25}} = \frac{19}{5} = 4 \times 5 = 20$$

$\div 5$

25) 95 (3
-75
$\boxed{20}$
(Rem)

$$\frac{\overset{7}{121} \times \overset{6}{215} \times \overset{-7}{31} \times \overset{4}{1904}}{19} = \frac{\overset{4}{42} \times \overset{-9}{-28}}{19} = -36 + 19 + 19 = 2$$

Rem

$\div 3$

Rem $\rightarrow 2 \times 3 = 6$ ✓

Test RanKING

1. **The remainder when 75 × 73 × 78 × 76 is divided by 34 is:**

75 × 73 × 78 × 76 को 34 से विभाजित करने पर प्राप्त शेषफल ज्ञात करें-

(a) 15
(b) ~~18~~
(c) 22
(d) 12

7 × 5 × 10 × 8   1 × 12

$$\frac{75 \times 73 \times 78 \times 76}{34} = \frac{35 \times 80}{34} = 12^{\text{Rem}}$$

2. **What is the remainder when the product of 335, 608 and 853 is divided by 13?**

335, 608 और 853 के गुणनफल को 13 से विभाजित करने पर प्राप्त शेषफल क्या होगा?

(a) 11 (b) 12

(c) 6 (d) 7

$(-3) \times (-3) \times (-5)$

$$\frac{335 \times 608 \times 853}{13} = \frac{-45}{13} = -(-7)$$

$= 7$

3. **Find the remainder:/शेषफल ज्ञात कीजिए:**

$$\frac{\overset{3}{1998} * \overset{4}{1999} * \overset{5}{2000}}{7} = ? \quad \frac{\cancel{60}}{7} = 4$$

4. Find the remainder:/शेषफल ज्ञात कीजिए:

$$\frac{\overset{-5}{132} * \overset{-2}{135} * \overset{-1}{136} * \overset{1}{138} * \overset{5}{142}}{137} = ?$$

-50 or $(-50+137)$

$= 87$

5. **Find the remainder:/शेषफल ज्ञात कीजिए:**

$$\frac{\overset{-3}{816}*\overset{+6}{825}*\overset{4}{823}*\overset{8}{827}}{819}=?\ -576 \text{ or } -576+819$$

$$= 243$$

6. **Find the remainder:/शेषफल ज्ञात कीजिए:**

$$\frac{\overset{2}{232} * \overset{-3}{43} * \overset{1}{2347} * \overset{2}{4671} * \overset{-4}{6988}}{23} = ?$$

$\frac{48}{23} = 2^{Rem}$

$\cancel{2300} + 47 +$

$\cancel{6900} + 88$

7. **Find the remainder when 49 × 51 × 54 × 37 × 123 is divided by 24.**

$$\frac{\overset{1}{49} \times \overset{3}{51} \times \overset{6}{54} \times \overset{13}{37} \times \overset{3}{123}}{24}$$

जब 49 × 51 × 54 × 37 × 123 को 24 से विभाजित किया जाता है तो शेषफल ज्ञात कीजिए।

(RRB RPF SI 2024)

A) 8 B) 4

C) 10 D) 6

$$\frac{\overset{6}{54} \times \overset{13}{13}}{24} = \cancel{78} \rightarrow 6$$

8. **What is the remainder when $70 \times 71 \times 72 \times 73 \times 74 \times 75 \times 76 \times 77 \times 78 \times 79$ is divided by 1000?**

जब 70×71×72×73×74×75×76×77×78×79 को 1000 से विभाजित किया जाता है तो शेषफल क्या होता है?

(UPSC CDS-2 2024)

(a) 3 (b) 2

(c) 1 (d) 0

125×8

$= 5^3 \times 2^3$

#

25×5

8

9. **What is the remainder when $91 \times 92 \times 93 \ldots\ldots\ldots .99$, is divided by 1261?**

जब 91×92×93.....97..×99 को 1261 से विभाजित किया जाता है, तो शेषफल क्या होगा ?

13

(a) 0 (b) 1

(c) 2 (d) 3

$1261 = 13 \times 97$

10. What is the smallest natural number n such that $(n+1) \times n \times (n-1) \times (n-2) \ldots \ldots .3 \times 2 \times 1$ is divisible by 910?

सबसे छोटी प्राकृतिक संख्या n क्या है जिससे $\boxed{(n+1) \times n \times (n-1) \times (n-2) \ldots\ldots 3 \times 2 \times 1}$, 910 से विभाज्य हो?

(CDS-1 2024)

(a) 91
(b) 90
(c) 13
(d) 12

= 13

$n_{smallest} = 12$

$910 \rightarrow 7 \times 10 \times 13$

Factorial

$1! = 1$

$2! = 2\times1 = 2$

$3! = 3\times2\times1 = 6$

$4! = 4\times3\times2\times1 = 24$

$5! = 5\times4\times3\times2\times1 = 120$

$6! = 6\times5! = 6\times120 = 720$

$7! = 7\times6! = 7\times720 = 5040$

$$n! = n\boxed{(n-1)(n-2)\times\cdots\times1}$$

$$n! = n\times(n-1)!$$

$$0! = 1$$

11. **Find the remainder:/शेषफल ज्ञात कीजिए:**

$1 + 2 + 6 + 24 + 0$

$$\frac{1!+2!+3!+4!+5!+---+880!}{120} = ?$$

Rem = 33